

Analysis First-Year Exam, January 2020, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let f be a bounded real-valued function on the set $X \times Y$. Prove that

$$\sup_{x \in X} \inf_{y \in Y} f(x, y) \leq \inf_{y \in Y} \sup_{x \in X} f(x, y).$$

Show by example that the inequality can be strict.

2. The diameter of the compact metric space X is defined by

$$\text{diam } X = \sup\{d(p, q) : p, q \in X\}.$$

Prove that there exist $p_0, q_0 \in X$ such that $\text{diam } X = d(p_0, q_0)$.

3. Let the function $f : X \rightarrow Y$ between metric spaces have closed graph $\{(x, y) : y = f(x)\} \subseteq X \times Y$. Prove the true and disprove the false:

- (i) if X is compact then f is continuous;
- (ii) if Y is compact then f is continuous.

4. Let $(s_n : n > 0)$ be a bounded real sequence and write $\sigma_n = (s_1 + \cdots + s_n)/n$ for each $n > 0$. Prove that

$$\limsup_n \sigma_n \leq \limsup_n s_n.$$

Hence, or otherwise, show that if $s_n \rightarrow s$ then $\sigma_n \rightarrow s$ also.

5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $|f(x) - f(a)| \leq A|x - a|^{1+\alpha}$ for some constants $A, \alpha > 0$ and all $a, x \in \mathbb{R}$. Prove that f is constant.